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10 CFR 50.73

May 23, 2019

Serial: RA-19-0244

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555

Subject: Brunswick Steam Electric Plant, Unit Nos 1 and 2
Renewed Facility Operating License Nos. DPR-71 and DPR-62
Docket No. 50-325 and 50-324
Licensee Event Report 1-2019-001

In accordance with the Code of Federal Regulations, Title 10, Part 50.73, Duke Energy Progress, LLC, submits the enclosed Licensee Event Report (LER). This report fulfills the requirement for a written report within sixty (60) days of a reportable occurrence.

This document contains no regulatory commitments.

Please refer any questions regarding this submittal to Mr. Jerry Pierce, Manager – Nuclear Support Services, at (910) 832-7931.

Sincerely,

A handwritten signature in blue ink, appearing to read "WRG", written over a light blue circular background.

William R. Gideon

TMS/tms

Enclosure: Licensee Event Report

cc (with enclosure):

U. S. Nuclear Regulatory Commission, Region II
ATTN: Ms. Catherine Haney, Regional Administrator
245 Peachtree Center Ave, NE, Suite 1200
Atlanta, GA 30303-1257

U. S. Nuclear Regulatory Commission
ATTN: Mr. Gale Smith, NRC Senior Resident Inspector
8470 River Road
Southport, NC 28461-8869

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11555 Rockville Pike
Rockville, MD 20852-2738

Chair - North Carolina Utilities Commission **(Electronic Copy Only)**
4325 Mail Service Center
Raleigh, NC 27699-4300
swatson@ncuc.net

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name

Brunswick Steam Electric Plant (BSEP), Unit 1

2. Docket Number

05000325

3. Page

1 OF 3

4. Title

Unplanned Automatic Start of Emergency Diesel Generators

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
03	25	2019	2019	- 001	- 00	05	23	2019	Brunswick Unit 2	05000324
									Facility Name	Docket Number
										05000

9. Operating Mode	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
100	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER**Licensee Contact**

Jerry Pierce, Manager – Nuclear Support Services

Telephone Number (Include Area Code)

(910) 832-7931

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
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14. Supplemental Report Expected☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No**15. Expected Submission Date**

Month	Day	Year

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

At 04:02 Eastern Daylight Time (EDT) on March 25, 2019, an actuation of the four Emergency Diesel Generators (EDGs) occurred. Unit 1 was in Mode 1 at approximately 100 percent power and Unit 2 was in Mode 4 at 0 percent power. Unit 2 was in the process of aligning the electrical distribution system to power the emergency buses via the Unit Auxiliary Transformer (UAT) when a main generator differential lockout relay actuated; thereby starting the EDGs. The emergency buses remained energized from offsite power via the Startup Auxiliary Transformer (SAT) and, therefore, the EDGs did not tie to their respective buses. The EDGs responded per design to this event.

The cause of event was a historical design change error. During replacement of the original Main Power Transformers (MPTs), the fact that the new transformers have lower second order harmonic in-rush current was not identified and, as a result, appropriate protective relay settings were not implemented.

Corrective actions include revising operating procedures to preclude EDG automatic starts on Main Generator primary lockout when restoring the UAT backfeed.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Brunswick Steam Electric Plant (BSEP), Unit 1	05000325	2019	- 001	- 00

NARRATIVE

Energy Industry Identification System (EIIIS) codes are identified in the text as [XX].

Background

At the time of the event, Unit 1 was in Mode 1 (i.e., Power Operation), at approximately 100 percent rated thermal power (RTP). Unit 2 was in Mode 4 (i.e. Cold Shutdown), at 0 percent RTP. There was no inoperable equipment that contributed to the event.

Reportability Criteria

This event is reportable per 10 CFR 50.73(a)(2)(iv)(A) because the plant experienced a valid start of its four Emergency Diesel Generators (EDGs) [EK]. The event was initially reported to the NRC on March 25, 2019 (i.e., Event Number 53955).

Event Description

At 04:02 Eastern Daylight Time (EDT) on March 25, 2019, an actuation of the EDGs occurred. Unit 2 was in the process of aligning the electrical distribution system to power the emergency buses via the Unit Auxiliary Transformer (UAT) [EA] when a main generator differential lockout relay actuated; thereby starting the EDGs. The emergency buses remained energized from offsite power via the Startup Auxiliary Transformer (SAT) and, therefore, the EDGs did not tie to their respective buses. The EDGs responded per design to this event. Troubleshooting and testing confirmed that the lockout was not a result of an actual fault condition or degraded equipment.

Event Cause

The direct cause of this event was a Main Generator primary lockout that occurred when generator breaker Power Circuit Breaker (PCB) 29A was closed during the alignment of the Unit 2 electrical distribution system to power the emergency buses via the UAT. The lockout occurred due to a trip of differential relay 2-GEN-87GT-2-A. The cause of event was a historical design change error.

The function of the differential relay is to identify a fault inside the zone of protection by comparing the currents flowing into the zone with the currents flowing out of the zone. If the difference is greater than a set percentage, the relay will actuate. The 87GT-2 relay compares the sum of the currents through PCBs 29A and 29B with the sum of the currents through the main generator and UAT. It is primarily designed to detect a fault in the main power transformer but the zone of protection also includes portions of the isophase bus and 230kV system. A difference in these currents indicates a fault. In the event, PCB 29B and the no-load disconnect switch were open, so the relay was comparing the current flow through PCB 29A with the current flow into the UAT.

When system voltage is applied to a transformer at a time when normal steady-state flux is at a different value from that existing in the transformer, a current transient occurs, known as magnetizing in-rush current. BSEP uses HU-1 relays for the differential protection. These relays have a differential unit, a harmonic restraint unit, and an indicating instantaneous trip unit. The differential unit and harmonic restraint unit contacts are in series. Per the vendor manual, the differential unit may or may not close its contact during

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Brunswick Steam Electric Plant (BSEP), Unit 1	05000325	2019	- 001	- 00

NARRATIVE

in-rush, depending on magnitude. The harmonic restraint contact will not close unless the second harmonic current is at least 15 percent of the fundamental component. Thus, prevention of spurious trips on in-rush is reliant on the in-rush current containing at least 15 percent second harmonics, which may not be the case with the relatively new main transformers. The original Main Power Transformers (MPTs) [EL] were replaced in 2004 for Unit 1 and 2005 for Unit 2. The harmonic restraint settings for the differential relays were not reduced to account for the lower second order harmonic in-rush current associated with new transformers.

The actual second harmonics of the event were at the point where the harmonic restraint on the differential relay could have actuated and blocked the trip (i.e., approximately 15 percent). Given the various tolerances inherently associated with electromechanical relays and sensing devices, it was concluded that the harmonics were not high enough to actuate the harmonic restraint and block the trip. This conclusion is supported by the relative age of the MPTs and the UATs. Had the harmonic restraint settings for the differential relays been reduced when the new transformers were installed, this event would not have occurred.

Safety Assessment

The safety significance of this event is minimal. The emergency buses remained energized from offsite power during the event and the EDGs responded per design to this event.

Corrective Actions

The corrective actions below are currently planned. Any changes to the corrective actions and schedules noted below will be made in accordance with the site's corrective action program.

- Revise procedures 1(2)OP-50, "Plant Electrical System Operating Procedure," to disable the EDG automatic start on Main Generator primary lockout prior to restoring UAT backfeed. This action is currently scheduled to be completed by October 3, 2019.

Previous Similar Events

A review of events for the past three years identified the following previous similar event where re-alignment of the electrical distribution systems resulted in actuation of the EDGs.

- Event Notification (EN) 52778, completed on May 31, 2017, reported an invalid actuation of the EDGs occurred when powering the 2C balance-of-plant (BOP) bus from the SAT. The event was a result of failure to install flexible links between the SAT and the 2D BOP bus. The corrective actions associated with EN 52778 could not have reasonably been expected to prevent the condition reported in LER 1-2019-001.

Commitments

No regulatory commitments are contained in this report.